

Radiative transfer theory for high-frequency power flows in fluid-saturated, poro-visco-elastic media

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Some recent mathematical results are used to study the propagation features of the high-frequency vibrational energy density in three-dimensional fluid-saturated, isotropic poro-visco-elastic media. The theory of high-frequency asymptotics of the solutions of hyperbolic partial differential equations shows that their energy satisfies Liouville-type transport or radiative transfer equations for randomly heterogeneous materials. For long propagation times these equations can be approached by diffusion equations. The corresponding diffusion parameters—mean-free paths and diffusion constants—associated with Biot's linear model for such media, are derived. The analysis accounts for the thermal and viscous memory effects of the fluid phase, and the viscous memory effect of the solid phase through time convolution operators. In this respect it also extends the existing mathematical results. © 2005 Acoustical Society of America. [DOI: 10.1121/1.1856271]

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I. INTRODUCTION

The theory of radiative transfer describes the high-frequency energy evolution in a linear random medium. It originates from considerations on the propagation of light through turbulent atmospheres.¹ In the frame of the classical multiple scattering theory of waves,² a radiative transfer equation for the intensity can be obtained from a Bethe–Salpeter (BS) equation under certain hypotheses, in particular when the wavelength is small compared to the macroscopic features of the medium, and perturbations of the latter are weak. In addition, the so-called ladder approximation is often invoked for the BS equation to make it suitable for analytical developments. However, it is only recently that a systematic derivation of transport equations for the energy in random electromagnetic, acoustic, or elastic media has been proposed.^{3–6} The analysis is based (i) on the use of a Wigner transform, the high-frequency, non-negative limit of which is the angularly resolved energy density in time and space; and (ii) on the introduction of an explicit scaling of wave propagation patterns. The mathematical framework, called microlocal analysis, is very general and applies to any kind of first-order hyperbolic partial differential equations. The purpose of this paper is to present how the theory applies to vibrations of fluid-saturated porous media as described by Biot's equations. The main objective is to derive the acoustic diffusion parameters for such materials, because they could be linked in some ways to their acoustic insulation and dissipation properties. Thermal dissipation and viscosity of the fluid as well as viscosity of the solid skeleton are taken into account in the analysis. This requires nontrivial extensions of the existing theory to the case of time convolution operators, as done in Ref. 7 for visco-elastic media. These results are adapted to poro-visco-elastic media in the present paper.

The scattering of a fast compression wave by a spherical inclusion in a fluid-saturated porous medium was considered in Ref. 8. These results are limited mostly to the lower frequency range and the case of a small Biot's modulus. In Ref.

9 this problem was re-examined for higher frequencies in the frame of a Born approximation. Waves scattered by an obstacle embedded in a porous medium were calculated by representational integral equations in Ref. 10, using *ad hoc* Green's kernels derived in the course of the paper. In Ref. 11 the author analyzed multiply scattered waves by perturbative expansions, but his work is limited to delta-correlated scatterers and by approximations inherent to perturbation techniques. None of these studies has considered the high-frequency transport and diffusion limits for arbitrarily random media as done here. Furthermore, this analysis is not reduced to a particular shape or distribution of scatterers, such as spherical or delta-correlated heterogeneities. The sole restriction is that random perturbations of the medium parameters shall be weak and isotropic, as effectively defined by the theory presented below. Also, one considers that they vary—either smoothly or not—at a length scale comparable to the wavelength, which is the *a priori* worst configuration since one can expect strong interactions between waves and heterogeneities of similar sizes.

In this paper, we first recall the basic mathematical tools we have used to describe the high-frequency energy density and its evolution with time for rapidly oscillating initial conditions. Then, this setting—extended to time convolution operators—is applied to Biot's equations describing the motion of a fluid-saturated, isotropic poro-visco-elastic medium. The effects of thermal dissipation at the interface between the fluid and solid phases and viscosity of the fluid are also considered. One concurrently obtains the exact analytical expressions of the compressional wave speeds in this regime. The influence of random perturbations of the mechanical parameters of the solid skeleton is analyzed in Sec. IV and the diffusion regime is depicted in Sec. V, where analytical expressions of the diffusion parameters, the transport mean free paths, and the diffusion constant are given. A numerical example is examined in Sec. VI using experimental data taken from the literature. Some conclusions are drawn in Sec. VII.

One notes that the same analysis of high-frequency energy propagation features has been proposed in Ref. 12, but calculations were not performed thoroughly and memory effects (viscosity and thermal coupling) were not taken into account.

II. THEORETICAL BACKGROUND: WIGNER MEASURES AS ENERGY DENSITY ESTIMATES

We summarize in this section the basic developments of Refs. 3–6 generalized to incorporate the time convolution operators in Ref. 7. The work reported in Ref. 7 deals with the extension of the theory to the case of a visco-elastic material with memory effects. The work reported here specializes it to the case of a poro-elastic medium that considers the thermal and viscous effects of the fluid phase. The interested reader may consult these references for a detailed presentation and the proofs associated with the results given below. The materials presented in Sec. II A and Sec. II B are basically adapted from Ref. 13.

Consider a first-order hyperbolic system of the form

$$\mathbf{A}(\mathbf{x}, 0) \partial_t \mathbf{v} + \sum_{j=1}^d \mathbf{D}_j \partial_{x_j} (\mathbf{v} + \mathbf{B}(\mathbf{x}, \cdot) *_{\epsilon} \mathbf{v}) + \mathbf{\Omega}(\mathbf{x}, \cdot) *_{\epsilon} \mathbf{v} = \mathbf{0}, \quad (1)$$

where $\mathbf{v}$ is the vector of state variables in $\mathbb{C}^n$, and $\mathbf{x} \in \mathbb{R}^d$ runs through the physical space; $d = 1, 2$, or 3 is its dimension, but for this application one lets $d = 3$ all along the paper. $\mathbf{A}(\mathbf{x}, t)$ is a symmetric, positive-definite $n \times n$ real matrix; $\mathbf{D}_j$ are symmetric $n \times n$ real matrices independent of position $\mathbf{x}$ and time t ; $\mathbf{B}(\mathbf{x}, t)$ is a symmetric $n \times n$ real matrix; $\mathbf{\Omega}(\mathbf{x}, t)$ is an $n \times n$ real matrix; and $*_{\epsilon}$ is the time convolution product $f *_{\epsilon} g(t) = \int_0^t f(\tau) g(t - \tau) d\tau$, $t \geq 0$. The solution of Eq. (1) is denoted by $\mathbf{v}^{\epsilon}$ for initial conditions oscillating rapidly (as detailed below; see Sec. II A) at an arbitrarily small scale ϵ , considering slowly varying mechanical coefficients such that neither $\mathbf{A}$ nor $\mathbf{B}$ depend on ϵ . One also assumes that the sequence $(\mathbf{v}^{\epsilon})$ is uniformly bounded in $(L^2(\mathbb{R}^3))^n$, where $L^2(\mathbb{R}^3)$ is the set of square integrable functions with respect to the space variable. Equation (1) is very general for wave equations arising from physical problems. It can embed Maxwell's equations for electromagnetic waves, the wave equation for acoustics, or Navier's equation of three-dimensional elastodynamics. For example, for an isotropic and synchronous visco-elastic medium characterized by its time-dependent Lamé's moduli $\lambda(\mathbf{x}, t)$ and $\mu(\mathbf{x}, t)$, and density $\varrho(\mathbf{x})$, the Navier equation of motion for the displacement vector $\mathbf{u} \in \mathbb{R}^3$ is

$$\varrho \partial_t^2 \mathbf{u} = \text{div } \boldsymbol{\sigma}(\mathbf{u}),$$

where the stress tensor $\boldsymbol{\sigma}$ is linked to the strain tensor $\boldsymbol{\epsilon}$ by Hooke's law

$$\boldsymbol{\sigma} = \lambda(\mathbf{x}, 0) \text{div } \mathbf{u} \mathbf{I}_3 + 2\mu(\mathbf{x}, 0) \boldsymbol{\epsilon} + \partial_t \lambda(\mathbf{x}, \cdot) *_{\epsilon} \text{div } \mathbf{u} \mathbf{I}_3 + 2\partial_t \mu(\mathbf{x}, \cdot) *_{\epsilon} \boldsymbol{\epsilon}.$$

Throughout the paper $\mathbf{I}_j$ is the $j \times j$ identity matrix. These equations have the form of Eq. (1), setting for example $\mathbf{v} = (\partial_t u_1, \partial_t u_2, \partial_t u_3, \epsilon_{11}, \epsilon_{22}, \epsilon_{33}, \epsilon_{23}, \epsilon_{13}, \epsilon_{12}, p)$ with $\epsilon_{ij} = \mu(\mathbf{x}, 0)(\partial_{x_j} u_i + \partial_{x_i} u_j)$ and $p = \lambda(\mathbf{x}, 0) \text{div } \mathbf{u}$, together with $\mathbf{A}(\mathbf{x}, t) = \text{diag}(\varrho(\mathbf{x}) \mathbf{I}_3, 1/2\mu(\mathbf{x}, t) \mathbf{I}_3, 1/\mu(\mathbf{x}, t) \mathbf{I}_3, 1/\lambda(\mathbf{x}, t))$

and $\mathbf{B}(\mathbf{x}, t) = \partial_t \mathbf{A}^{-1}(\mathbf{x}, t) \mathbf{A}(\mathbf{x}, 0)$. Other definitions of $\mathbf{v}$ and $\mathbf{A}$ which lead to the same general form are obviously possible, as in Ref. 14, where the Navier equation of elastodynamics in $\mathbb{R}^d$ is written in the form of a symmetric hyperbolic system (1) with $n = d + 1$.

A. Energy and power flow estimates

One injects high-frequency energy into the system described by Eq. (1) through, say, initial conditions of the form

$$\mathbf{v}^{\epsilon}(\mathbf{x}, 0) = \mathbf{v}_0^{\epsilon}(\mathbf{x}), \quad (2)$$

where the data $\mathbf{v}_0^{\epsilon}$ are bounded in $(L^2(\mathbb{R}^3))^n$ as ϵ goes to 0 and oscillating at most at the frequency $1/\epsilon$. For example, one can take $\mathbf{v}_0^{\epsilon}(\mathbf{x}) = \mathbf{v}_0(\mathbf{x}) e^{i\epsilon \mathbf{x} / \epsilon}$, where $\mathbf{v}_0$ as well as its derivatives are in $(L^2(\mathbb{R}^3))^n$, or $\mathbf{v}_0^{\epsilon}(\mathbf{x}) = \epsilon^{-3/2} \mathbf{v}_0(\mathbf{x}/\epsilon)$, where $\mathbf{v}_0$ is such that $\|\nabla \mathbf{v}_0\|_{L^2} < +\infty$, but more general conditions to be fulfilled by the initial data for the following to apply are given in Refs. 3 and 4. The energy density associated with the solution $\mathbf{v}^{\epsilon}$ of Eqs. (1)–(2) is defined by

$$\mathcal{E}^{\epsilon}(\mathbf{x}, t) = \frac{1}{2} (\mathbf{v}^{\epsilon}, \mathbf{v}^{\epsilon})_{\mathbf{A}(\mathbf{x}, 0)}, \quad (3)$$

while the densities of power flow along each direction $j = 1, 2$, or 3 are

$$\Pi_j^{\epsilon}(\mathbf{x}, t) = \frac{1}{2} (\mathbf{v}^{\epsilon}, \mathbf{v}^{\epsilon})_{\mathbf{D}_j}, \quad (4)$$

with the notation $(\mathbf{u}, \mathbf{v})_{\mathbf{M}} := (\mathbf{M} \mathbf{u}, \bar{\mathbf{v}})$, where $\bar{\mathbf{v}}$ is the complex conjugate of $\mathbf{v}$. A potential good candidate to study the high-frequency energetic features of the system (1)–(2) is the Wigner transform, because it has a direct link with the above quantities in the limit $\epsilon \rightarrow 0$, as well as some other nice evolution properties given below (see Sec. II B). Let us define for two functions $\mathbf{u}$ and $\mathbf{v}$ lying in a bounded subset of $(L^2(\mathbb{R}^3))^n$

$$\mathbf{W}^{\epsilon}(\mathbf{u}, \mathbf{v})(\mathbf{x}, \mathbf{k}, t) = \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} e^{i\mathbf{k} \cdot \mathbf{y}} \mathbf{u} \left(\mathbf{x} - \frac{1}{2} \epsilon \mathbf{y}, t \right) \overline{\mathbf{v} \left(\mathbf{x} + \frac{1}{2} \epsilon \mathbf{y}, t \right)} d\mathbf{y}. \quad (5)$$

Then, the $n \times n$ matrix-valued Wigner transform $\mathbf{W}^{\epsilon}[\mathbf{v}^{\epsilon}] := \mathbf{W}^{\epsilon}(\mathbf{v}^{\epsilon}, \mathbf{v}^{\epsilon})$ of $\mathbf{v}^{\epsilon}$ has (up to an extracted subsequence) a Hermitian limit $\mathbf{W}^{\epsilon}[\mathbf{v}^{\epsilon}] \rightarrow \mathbf{W}$ as $\epsilon \rightarrow 0$ in the sense of tempered distributions which is also a non-negative measure,^{3,4} the so-called Wigner measure of $\mathbf{v}^{\epsilon}$.

Assuming that $\mathbf{v}^{\epsilon}$ of Eqs. (1)–(2) has the same oscillatory properties as the initial conditions, it can be shown, owing to Proposition 1.7 of Ref. 4 or Proposition 2.3.3 of Ref. 7, that the energy $\mathcal{E}(\mathbf{x}, t)$ and power flow $\Pi_j(\mathbf{x}, t)$ densities in the limit $\epsilon \rightarrow 0$ are estimated from the Wigner measure by

$$\lim_{\epsilon \rightarrow 0} \mathcal{E}^{\epsilon}(\mathbf{x}, t) := \mathcal{E}(\mathbf{x}, t) = \frac{1}{2} \int_{\mathbb{R}^3} \mathbf{A}(\mathbf{x}, 0) : \mathbf{W}(\mathbf{x}, \mathbf{k}, t) d\mathbf{k}, \quad (6)$$

and

$$\lim_{\epsilon \rightarrow 0} \Pi_j^{\epsilon}(\mathbf{x}, t) := \Pi_j(\mathbf{x}, t) = \frac{1}{2} \int_{\mathbb{R}^3} \mathbf{D}_j : \mathbf{W}(\mathbf{x}, \mathbf{k}, t) d\mathbf{k}, \quad (7)$$

for $j=1, 2, 3$, respectively, where $\mathbf{A}:\mathbf{B}=\text{Tr}(\mathbf{A}\mathbf{B}^T)$ is the usual tensor scalar product and Tr is the trace of a matrix.

B. Transport properties of Wigner measures

One now turns to the evolution properties of the Wigner measure associated with the sequence $(\mathbf{u}^\epsilon)$ of solutions of the system (1)–(2). Define $[\mathbf{A}, \mathbf{B}] = \mathbf{A}\mathbf{B} + \mathbf{B}^* \mathbf{A}^*$, where $\mathbf{A}^* = \mathbf{A}^T$ stands for the adjoint of a matrix. The Wigner measure satisfies the dispersion equation

$$[\mathbf{iL}, \mathbf{W}] = \mathbf{0}, \quad (8)$$

where $\mathbf{L}$ is the dispersion matrix of the above system defined by

$$\mathbf{L}(\mathbf{x}, \mathbf{k}) = \mathbf{A}^{-1}(\mathbf{x}, 0) \sum_{j=1}^3 k_j \mathbf{D}_j. \quad (9)$$

It is self-adjoint with respect to the scalar product $(\mathbf{u}, \mathbf{v})_{\mathbf{A}(\mathbf{x}, 0)}$; thus, its n right eigenvectors form an orthonormal family with respect to this scalar product. The latter and the associated real eigenvalues are denoted by $\mathbf{b}_{\alpha_r}(\mathbf{x}, \mathbf{k})$ and $\omega_{\alpha_r}(\mathbf{x}, \mathbf{k})$, respectively, for $1 \leq \alpha \leq m \leq n$ and $1 \leq r \leq R_\alpha$, where R_α is the order of multiplicity of eigenvalue ω_α for a mode α among the m existing ones, such that $\sum_{\alpha=1}^m R_\alpha = n$. R_α is assumed to be independent of $(\mathbf{x}, \mathbf{k})$. From the dispersion equation the Wigner measure may be written

$$\mathbf{W}(\mathbf{x}, \mathbf{k}, t) = \sum_{\alpha=1}^m \sum_{r,s=1}^{R_\alpha} w_{\alpha}^{rs}(\mathbf{x}, \mathbf{k}, t) \mathbf{b}_{\alpha_r}(\mathbf{x}, \mathbf{k}) \otimes \mathbf{b}_{\alpha_s}(\mathbf{x}, \mathbf{k}), \quad (10)$$

where the w_{α}^{rs} 's are scalar coefficients, hereafter referred to as specific intensities. If one defines coherence matrices $\mathbf{W}_\alpha$, $1 \leq \alpha \leq m$, by

$$\mathbf{W}_{\alpha,rs}(\mathbf{x}, \mathbf{k}, t) = w_{\alpha}^{rs}(\mathbf{x}, \mathbf{k}, t), \quad 1 \leq r, s \leq R_\alpha, \quad (11)$$

then they will satisfy Liouville-type transport equations

$$\partial_t \mathbf{W}_\alpha + \{\omega_\alpha, \mathbf{W}_\alpha\} + [\mathbf{W}_\alpha, \mathbf{N}_\alpha] = \mathbf{0}. \quad (12)$$

$\{ \}$ represents the usual Poisson's bracket defined by $\{f, g\} = \nabla_{\mathbf{k}} f \cdot \nabla_{\mathbf{x}} g - \nabla_{\mathbf{x}} f \cdot \nabla_{\mathbf{k}} g$ and the $\mathbf{N}_\alpha$'s are skew-symmetric matrices given by

$$N_{\alpha,rs} = \sum_{j=1}^3 (\partial_{x_j} \mathbf{b}_{\alpha_r}, \mathbf{b}_{\alpha_s}) \mathbf{D}_j - (\nabla_{\mathbf{x}} \omega_\alpha \cdot \nabla_{\mathbf{k}} \mathbf{b}_{\alpha_r}, \mathbf{b}_{\alpha_s})_{\mathbf{A}(\mathbf{x}, 0)} - \frac{1}{2} \nabla_{\mathbf{k}} \cdot \nabla_{\mathbf{x}} \omega_\alpha \delta_{rs}, \quad (13)$$

for $1 \leq r, s \leq R_\alpha$. These rotation matrices arise from the curvature of rays in heterogeneous media, leading to rotation of the polarization vector around a ray as waves propagate.⁵ It is understood that if $R_\alpha = 1$ the last term in Eq. (12) always vanishes and the latter writes

$$\partial_t \mathbf{W}_\alpha + \{\omega_\alpha, \mathbf{W}_\alpha\} = \mathbf{0}, \quad (14)$$

dropping subscripts r and s . These results show that the transport equations satisfied by specific intensities are basically unchanged from the case without time-convolution operators.⁷

III. TRANSPORT EQUATIONS FOR ISOTROPIC POROUS MEDIA

Elastic waves in a fluid-saturated porous solid are described by the model of Ref. 15, taking into account the approach of Ref. 16 as well. The macroscopically homogeneous and isotropic porous material is constituted by a solid phase and a fluid phase which saturates the open porous space. The solid phase includes a solid part constituted by a synchronous (Poisson's coefficient ν is independent of time) visco-elastic material and the closed porosity. Both phases have average displacements denoted by $\mathbf{u}_s$ and $\mathbf{u}_f$ for the solid and fluid ones, respectively. The averaging process corresponds to the homogenization at a macroscopic scale of the quantities observed at the microscopic scale of the medium where they are not necessarily continuous and materials are heterogeneous. The ratio of the volume occupied by the fluid phase with respect to the overall volume of the porous medium is given by its porosity Φ such that $0 \leq \Phi < 1$. The density of the porous material is $\varrho = (1 - \Phi)\varrho_s + \Phi\varrho_f$, where ϱ_s and ϱ_f are the densities of the materials constituting the solid and fluid phases, respectively. One assumes that Φ , ϱ_s , and ϱ_f are constant over the space. The average fluid motion with respect to the solid phase is given by the filtration velocity $\dot{\mathbf{w}} = \Phi(\dot{\mathbf{u}}_f - \dot{\mathbf{u}}_s)$, where $\dot{\mathbf{u}}$ is the time derivative of $\mathbf{u}$ under the assumption of small transformations of the medium. The latter occupies an open domain Ω of $\mathbb{R}^3$.

A. Basic equations

The constitutive equations are written¹⁶ for the porous material

$$\boldsymbol{\sigma}^{\text{tot}} = (\lambda_0 + \partial_t \lambda^*) \nabla \cdot \mathbf{u}_s \mathbf{I}_3 + 2(\mu_0 + \partial_t \mu^*) \nabla \otimes_s \mathbf{u}_s - p \mathcal{B}, \quad (15)$$

and for the fluid phase

$$p + \beta^* p = -M(\mathcal{B} : \nabla \otimes_s \mathbf{u}_s + \nabla \cdot \mathbf{w}). \quad (16)$$

The pore fluid pressure is p , $\boldsymbol{\sigma}^{\text{tot}}$ is the total macroscopic stress tensor of the porous medium, λ and μ are the time-dependent Lamé's moduli of the solid phase with $\lambda_0 = \lambda(0)$ and $\mu_0 = \mu(0)$, β is a time-dependent compressibility coefficient of the fluid phase which accounts for thermal effects, M is Biot's modulus, and B is the coupling coefficient between phases such that the coupling tensor is $\mathcal{B} = B \mathbf{I}_3$ in the isotropic case. By definition¹⁶

$$\frac{1}{M} = \frac{\Phi}{K_f} + \frac{B - \Phi}{K_s}, \quad B = 1 - \frac{K_b}{K_s},$$

where K_f is the fluid bulk modulus, $K_b = \lambda_0 + (2/3)\mu_0$ is the macroscopic bulk modulus of the dry solid phase, whereas K_s is the microscopic bulk modulus of the solid grains constituting the skeleton of the porous material such that $K_s \geq K_b$. Last, $\otimes_s$ is the symmetrized tensor product. Let

$$q = \lambda_0 \nabla \cdot \mathbf{u}_s - Bp, \quad (17a)$$

$$\boldsymbol{\epsilon} = 2\mu_0 \nabla \otimes_s \mathbf{u}_s, \quad (17b)$$

then

$$\nabla \cdot \dot{\mathbf{u}}_s = \frac{1}{\lambda_0} \dot{q} + \frac{B}{\lambda_0} \dot{p}, \quad (18a)$$

$$\nabla \cdot \dot{\mathbf{w}} = -\frac{B}{\lambda_0} \dot{q} - \left(\frac{1}{M} + \frac{B^2}{\lambda_0} \right) \dot{p} - \frac{\beta}{M} p. \quad (18b)$$

The dynamic equation of motion of the bulk mixture is written

$$\nabla \left(\boldsymbol{\epsilon} + \frac{\partial_t \mu}{\mu_0} * \boldsymbol{\epsilon} \right) + \nabla \left(q + \frac{\partial_t \lambda}{\lambda_0} * (q + Bp) \right) = \varrho \ddot{\mathbf{u}}_s + \varrho_f \ddot{\mathbf{w}}. \quad (19)$$

The linearized Darcy's law under the assumption of small transformations of the homogeneous medium is written

$$-\eta_f \dot{\varphi} \mathcal{K}^{-1} * \dot{\mathbf{w}} = \nabla p + \varrho_f (\alpha \ddot{\mathbf{u}}_f + (1 - \alpha) \ddot{\mathbf{u}}_s), \quad (20)$$

and describes the motion of the pore fluid relative to the porous skeleton. In the above equation, η_f is the pore fluid viscosity, φ is a time-dependent viscosity correction factor, and $\mathcal{K}$ is the intrinsic permeability tensor which depends on the geometry of the pore network solely; it is spherical in the isotropic case $\mathcal{K} = \mathcal{K} \mathbf{I}_3$, where $\mathcal{K}$ is the coefficient of permeability. At last, α is the tortuosity defined as the ratio of the microscopic mean-square velocity of fluid particles with re-

spect to the skeleton to their macroscopic quadratic velocity $|\dot{\mathbf{u}}_f - \dot{\mathbf{u}}_s|^2$. It is a purely geometrical quantity independent of fluid or solid densities such that $\alpha \geq 1$. Regarding the introduction of convolution products in Eqs. (15), we refer the reader to Fung's book¹⁷ for a detailed presentation of the behavior of viscous materials including memory effects. Regarding the introduction of convolution products in Eqs. (16) and (20), we refer to the work of Fellah and Depollier,¹⁸ extending their results to the case of a nonrigid solid frame. In this study, φ and β are proportional to the function $t \mapsto 1/\sqrt{t}$ for $t \downarrow 0$ (singular memory kernels). Note that Darcy's law may also be written $-\mathbf{Y} \dot{\mathbf{w}} = \nabla p + \varrho_f \ddot{\mathbf{u}}_s$, where $\mathbf{Y} := \eta_f \dot{\varphi} \mathcal{K}^{-1} * \mathbf{I} + (\alpha \varrho_f / \Phi) \partial_t$ is the viscodynamic operator introduced in Ref. 19.

Now, Eqs. (17) through (20) can be recast in the form of a first-order hyperbolic system of Eq. (1) with $n = 14$ and $d = 3$, introducing the state variable $\mathbf{v} = (\dot{\mathbf{u}}_s, \dot{\mathbf{w}}, \boldsymbol{\epsilon}, q, -p)$ of $\mathbb{C}^{14}$, where $\boldsymbol{\epsilon}$ is rearranged in a vector form as $\boldsymbol{\epsilon} = (\epsilon_{11}, \epsilon_{22}, \epsilon_{33}, \epsilon_{23}, \epsilon_{13}, \epsilon_{12})$. This leads to the equation

$$\mathbf{A}_0 \partial_t \mathbf{v} + \sum_{j=1}^3 \mathbf{D}_j \partial_{x_j} (\mathbf{v} + \mathbf{B} * \mathbf{v}) + \mathbf{\Omega} * \mathbf{v} = \mathbf{0}, \quad (21)$$

where $\mathbf{A}_0 = \mathbf{A}(0)$, with

$$\mathbf{A}(t) = \begin{bmatrix} \varrho \mathbf{I}_3 & \varrho_f \mathbf{I}_3 & 0 & 0 & 0 & 0 \\ \varrho_f \mathbf{I}_3 & \frac{\alpha \varrho_f}{\Phi} \mathbf{I}_3 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2\mu(t)} \mathbf{I}_3 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{\mu(t)} \mathbf{I}_3 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{\lambda(t)} & -\frac{B}{\lambda(t)} \\ 0 & 0 & 0 & 0 & -\frac{B}{\lambda(t)} & \frac{1}{M} + \frac{B^2}{\lambda(t)} \end{bmatrix},$$

$\mathbf{B} = \partial_t (\mathbf{A}^{-1}) \mathbf{A}_0$ and a diagonal matrix $\mathbf{\Omega}$ given by $\mathbf{\Omega} = \text{diag}(0, \eta_f \dot{\varphi} \mathcal{K}^{-1}, 0, 0, 0, \beta/M)$. Matrices $\mathbf{D}_1$, $\mathbf{D}_2$, and $\mathbf{D}_3$ are symmetric and have the form

$$-\mathbf{D}_j = \begin{bmatrix} 0 & 0 & \mathbf{K}_j & \mathbf{M}_j & \mathbf{k}_j & 0 \\ & 0 & 0 & 0 & 0 & \mathbf{k}_j \\ & & 0 & 0 & 0 & 0 \\ & & & 0 & 0 & 0 \\ & & & & 0 & 0 \\ & & & & & 0 \end{bmatrix},$$

where $\mathbf{K}_j$ and $\mathbf{M}_j$ are the 3×3 matrices given by their elements $K_j^{rs} = \delta_{jr} \delta_{js}$ and $M_j^{rs} = (1 - \delta_{jr})(1 - \delta_{js})(1 - \delta_{rs})$, and

$\mathbf{k}_j$ is the vector given by its components $k_j^r = \delta_{jr}$, with $r, s \in \{1, 2, 3\}$ and δ_{rs} being the Kronecker symbol. The results of the previous section are applied to this system under the assumption that the wavelength, scaled by ϵ , remains much larger than the pore and solid grain sizes in order to ensure the linearity of the proposed model. One also considers “ ϵ -oscillating” initial data $\mathbf{v}(\mathbf{x}, 0) = \mathbf{v}_0^\epsilon(\mathbf{x})$ with $\mathbf{v} = \mathbf{0}$ for $t < 0$.

B. High-frequency energy transport properties

The dispersion matrix for a porous medium thus writes

$$-\mathbf{L}(\mathbf{k}) = \begin{bmatrix} 0 & 0 & \frac{1}{\varrho_{sf}} \mathbf{K}(\mathbf{k}) & \frac{1}{\varrho_{sf}} \mathbf{M}(\mathbf{k}) & \frac{1}{\varrho_{sf}} \mathbf{k} & -\frac{\Phi}{\alpha \varrho_{sf}} \mathbf{k} \\ 0 & 0 & -\frac{\Phi}{\alpha \varrho_{sf}} \mathbf{K}(\mathbf{k}) & -\frac{\Phi}{\alpha \varrho_{sf}} \mathbf{M}(\mathbf{k}) & -\frac{\Phi}{\alpha \varrho_{sf}} \mathbf{k} & \frac{\Phi \varrho}{\alpha \varrho_{sf} \varrho_f} \mathbf{k} \\ 2\mu_0 \mathbf{K}(\mathbf{k}) & 0 & 0 & 0 & 0 & 0 \\ \mu_0 \mathbf{M}(\mathbf{k}) & 0 & 0 & 0 & 0 & 0 \\ (\lambda_0 + MB^2) \mathbf{k}^T & MB \mathbf{k}^T & 0 & 0 & 0 & 0 \\ MB \mathbf{k}^T & M \mathbf{k}^T & 0 & 0 & 0 & 0 \end{bmatrix}, \quad (22)$$

where $\alpha \varrho_{sf} = \alpha(1 - \Phi) \varrho_s + (\alpha - 1) \Phi \varrho_f$, $\mathbf{k} = (k_1, k_2, k_3)$, and

$$\mathbf{K}(\mathbf{k}) = \sum_{j=1}^3 k_j \mathbf{K}_j = \text{diag}(k_1, k_2, k_3),$$

$$\mathbf{M}(\mathbf{k}) = \sum_{j=1}^3 k_j \mathbf{M}_j = \begin{bmatrix} 0 & k_3 & k_2 \\ k_3 & 0 & k_1 \\ k_2 & k_1 & 0 \end{bmatrix}.$$

Let $|\mathbf{k}| = \sqrt{k_1^2 + k_2^2 + k_3^2}$ and $\hat{\mathbf{k}} = \mathbf{k}/|\mathbf{k}|$. One introduces vectors $\hat{\mathbf{z}}_1(\mathbf{k})$ and $\hat{\mathbf{z}}_2(\mathbf{k})$ such that $(\hat{\mathbf{k}}, \hat{\mathbf{z}}_1, \hat{\mathbf{z}}_2)$ form an orthonormal triplet, for example $\mathbf{k} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$, $\hat{\mathbf{z}}_1(\mathbf{k}) = (\cos \theta \cos \phi, \cos \theta \sin \phi, -\sin \theta)$ and $\hat{\mathbf{z}}_2(\mathbf{k}) = (-\sin \phi, \cos \phi, 0)$ in spherical coordinates. Then, the eigenvalues of the dispersion matrix in the set $E = \mathbb{R}^3 \times \mathbb{R}^3 \setminus \{(\mathbf{x}, \mathbf{k}); \mathbf{k} = \mathbf{0}\}$ are

$$\begin{aligned} \omega_0 &= 0 \quad \text{with multiplicity } 6, \\ \omega_S^\pm &= \pm c_S |\mathbf{k}| \quad \text{each with multiplicity } 2, \\ \omega_{p1}^\pm &= \pm c_{p1} |\mathbf{k}| \quad \text{each with multiplicity } 1, \\ \omega_{p2}^\pm &= \pm c_{p2} |\mathbf{k}| \quad \text{each with multiplicity } 1, \end{aligned} \quad (23)$$

where c_{p1}^2 and c_{p2}^2 are the solutions of

$$X^2 - (c_p^2 + c_1^2)X + c_p^2 c_2^2 = 0, \quad (24)$$

with parameters $c_S^2 = \mu_0 / \varrho_{sf}$, $c_p^2 = (\lambda_0 + 2\mu_0) / \varrho_{sf}$, and

$$\begin{aligned} c_1^2 &= \frac{M}{\varrho_{sf}} \left[B^2 + \frac{\Phi}{\alpha} \left(\frac{\varrho - 2B\varrho_f}{\varrho_f} \right) \right], \\ c_2^2 &= \frac{\Phi M}{\alpha \varrho_f} = c_1^2 - \frac{M}{\varrho_{sf}} \left(B - \frac{\Phi}{\alpha} \right)^2. \end{aligned}$$

These results agree with those of Refs. 20 and 21. One chooses $c_{p2} < c_{p1}$ in accordance with the usual notations of the dedicated literature. Therefore, $m=7$ and $R_0=6$, $R_{S+} = R_{S-} = 2$ with the notations of Sec. II B. The associated eigenvectors on E are

$$\begin{aligned} \mathbf{b}_r &= \left(0, 0, \sqrt{2\mu_0} \mathbf{K}(\hat{\mathbf{z}}_r) \hat{\mathbf{z}}_r, \sqrt{\frac{\mu_0}{2}} \mathbf{M}(\hat{\mathbf{z}}_r) \hat{\mathbf{z}}_r, 0, 0 \right), \\ \mathbf{b}_3 &= (0, 0, 2\sqrt{\mu_0} \mathbf{K}(\hat{\mathbf{z}}_1) \hat{\mathbf{z}}_2, \sqrt{\mu_0} \mathbf{M}(\hat{\mathbf{z}}_1) \hat{\mathbf{z}}_2, 0, 0), \end{aligned}$$

$$\begin{aligned} \mathbf{b}_4 &= \left(0, 0, \sqrt{\frac{2\lambda_0\mu_0}{\lambda_0 + 2\mu_0}} \mathbf{K}(\hat{\mathbf{k}}) \hat{\mathbf{k}}, \sqrt{\frac{\lambda_0\mu_0}{2(\lambda_0 + 2\mu_0)}} \mathbf{M}(\hat{\mathbf{k}}) \hat{\mathbf{k}}, \right. \\ &\quad \left. - \sqrt{\frac{2\lambda_0\mu_0}{\lambda_0 + 2\mu_0}}, 0 \right), \end{aligned} \quad (25)$$

$$\mathbf{b}_{4+r} = \left(0, \sqrt{\frac{\Phi}{\alpha \varrho_f}} \hat{\mathbf{z}}_r, 0, 0, 0, 0 \right),$$

$$\begin{aligned} \mathbf{b}_{Sr}^\pm &= \left(\frac{\hat{\mathbf{z}}_r}{\sqrt{2\varrho_{sf}}}, -\frac{\Phi \hat{\mathbf{z}}_r}{\alpha \sqrt{2\varrho_{sf}}}, \mp \sqrt{2\mu_0} \mathbf{K}(\hat{\mathbf{k}}) \hat{\mathbf{z}}_r, \right. \\ &\quad \left. \mp \sqrt{\frac{\mu_0}{2}} \mathbf{M}(\hat{\mathbf{k}}) \hat{\mathbf{z}}_r, 0, 0 \right), \end{aligned}$$

$$\begin{aligned} \mathbf{b}_{pj}^\pm &= \left(\frac{\hat{\mathbf{k}}}{\sqrt{2\varrho_{sf}}}, -\frac{\Phi \hat{\mathbf{k}}}{\alpha \sqrt{2\varrho_{sf}}}, \mp \frac{2\mu_0 \mathbf{K}(\hat{\mathbf{k}}) \hat{\mathbf{k}}}{c_{pj} \sqrt{2\varrho_{sf}}}, \mp \frac{\mu_0 \mathbf{M}(\hat{\mathbf{k}}) \hat{\mathbf{k}}}{c_{pj} \sqrt{2\varrho_{sf}}}, \right. \\ &\quad \left. \mp \frac{\lambda_0 + MB \left(B - \frac{\Phi}{\alpha} \right)}{c_{pj} \sqrt{2\varrho_{sf}}}, \mp \frac{M \left(B - \frac{\Phi}{\alpha} \right)}{c_{pj} \sqrt{2\varrho_{sf}}} \right), \end{aligned}$$

for $r, j \in \{1, 2\}$. Eigenvectors $\mathbf{b}_5$ and $\mathbf{b}_6$ correspond to acoustic transverse advection modes of the fluid phase, and are similar to the nonpropagating modes obtained for a purely acoustic medium.⁵ Eigenmodes $\mathbf{b}_1$ through $\mathbf{b}_4$ are their counterparts for a purely elastic medium. The main changes introduced by porosity compared to the elastic case are (i) the emergence of two compressional modes, instead of a single one, and (ii) the introduction of an effective density ϱ_{sf} in the definition of the shear and compression modes, which are similar to the elastic modes omitting this modification. The purely elastic medium is recovered if one lets $\Phi=0$ and $\alpha=1$, as one also has $B=0$ in this limit. Finally, one notes that these modes are nondispersive in accordance with some experimental observations done for two different porous materials (plastic foams).²²

Regarding the slow-wave speed c_{p2} , it is null for $\Phi=0$ and its value for $\Phi \neq 0$ and $\varrho_f / \varrho_s \ll 1$ or $K_f / K_b \ll 1$ approaches the modified velocity of sound in the air filling the pores $c_f = \sqrt{K_f / \alpha \varrho_f} = c / \sqrt{\alpha}$, in accordance with the high-frequency asymptotic estimate given in Refs. 20 and 21. Thus, one has $c_{p2} \leq c$. For such ratios of fluid to solid phase

bulk moduli, it may be observed that the coupling coefficient B has almost no influence and could be fixed at 1.

Now, let $w_{pj}^{\pm}(\mathbf{x}, \mathbf{k}, t)$, $j \in \{1, 2\}$, be the specific intensities for compressional waves corresponding to eigenvectors $\mathbf{b}_{pj}^{\pm}(\mathbf{k})$. They then satisfy the following transport equations:

$$\partial_t w_{pj}^{\pm} \pm c_{pj} \hat{\mathbf{k}} \cdot \nabla_{\mathbf{x}} w_{pj}^{\pm} = 0, \quad j \in \{1, 2\}. \quad (26)$$

Let $\mathbf{W}_S^{\pm}(\mathbf{x}, \mathbf{k}, t)$ be the 2×2 coherence matrices of specific intensities corresponding to eigenvectors $\mathbf{b}_{Sr}^{\pm}(\mathbf{k})$, $r \in \{1, 2\}$. Thus, they satisfy the following transport equations:

$$\partial_t \mathbf{W}_S^{\pm} \pm c_S \hat{\mathbf{k}} \cdot \nabla_{\mathbf{x}} \mathbf{W}_S^{\pm} = 0, \quad (27)$$

since rotation matrix $\mathbf{N}_S$ is null for homogeneous parameters, and the contributions of the convolution operator Ω cancel out in the limit $\epsilon \rightarrow 0$. Finally, the 6×6 coherence matrix $\mathbf{W}_0(\mathbf{x}, \mathbf{k}, t)$ satisfies $\partial_t \mathbf{W}_0 = 0$, since here again $\mathbf{N}_0 = 0$; thus, $\mathbf{W}_0(\mathbf{x}, \mathbf{k}, t) = 0$ if it is initially null. One observes that $w_{pj}^{\pm}(\mathbf{x}, \mathbf{k}, t) = w_{pj}^{\pm}(\mathbf{x}, -\mathbf{k}, t)$, $j \in \{1, 2\}$, and $\mathbf{W}_S^{\pm}(\mathbf{x}, \mathbf{k}, t) = \mathbf{W}_S^{\pm}(\mathbf{x}, -\mathbf{k}, t)$, so that the energy density is estimated by

$$\begin{aligned} \mathcal{E}(\mathbf{x}, t) = & \sum_{j=1,2} \int_{\mathbb{R}^3} w_{pj}^+(\mathbf{x}, \mathbf{k}, t) d\mathbf{k} + \int_{\mathbb{R}^3} \text{Tr} \mathbf{W}_S^+(\mathbf{x}, \mathbf{k}, t) d\mathbf{k} \\ & + \frac{1}{2} \int_{\mathbb{R}^3} \text{Tr} \mathbf{W}_0(\mathbf{x}, \mathbf{k}, t) d\mathbf{k}, \end{aligned} \quad (28)$$

while the power flow density vector is estimated by

$$\begin{aligned} \Pi(\mathbf{x}, t) = & \sum_{j=1,2} c_{pj} \int_{\mathbb{R}^3} w_{pj}^+(\mathbf{x}, \mathbf{k}, t) \hat{\mathbf{k}} d\mathbf{k} \\ & + c_S \int_{\mathbb{R}^3} \text{Tr} \mathbf{W}_S^+(\mathbf{x}, \mathbf{k}, t) \hat{\mathbf{k}} d\mathbf{k}, \end{aligned} \quad (29)$$

since the modes $\mathbf{b}_1(\mathbf{k})$ through $\mathbf{b}_6(\mathbf{k})$ do not propagate.

IV. THE RANDOM CASE: RADIATIVE TRANSFER

A. General considerations

In this section one considers some randomness of the mechanical properties of the medium and follows again the developments of Refs. 5 and 6. In their work these authors introduce a random perturbation $(\mathbf{V}(\mathbf{x}), \mathbf{x} \in \mathbb{R}^3)$ of matrix $\mathbf{A}$ of the form

$$\mathbf{A}(\mathbf{x}) = \mathbf{A}(\mathbf{x}) \left[\mathbf{I}_n + \sqrt{\epsilon} \mathbf{V} \left(\frac{\mathbf{x}}{\epsilon} \right) \right]. \quad (30)$$

The magnitude $\sqrt{\epsilon}$ of perturbations is chosen such that they have comparable effects on the unperturbed background medium as characterized by the mean matrix of parameters $\mathbf{A}(\mathbf{x})$. However, they need not be too strong, for in this case localization rather than transport may occur. Their correlation lengths are chosen to be comparable to wavelengths ϵ to allow strong interactions between the waves and the heterogeneities. For the system (1) to remain symmetric hyperbolic, the perturbation matrix has to satisfy the condition $\mathbf{A}(\mathbf{x})\mathbf{V}(\mathbf{y}) = \mathbf{V}^*(\mathbf{y})\mathbf{A}(\mathbf{x})$ for all $\mathbf{x}$ and $\mathbf{y}$ of $\mathbb{R}^3$. Parameter fluctuations are modeled with a statistically homogeneous, second-order matrix-valued random process with mean zero whose covariance functions are denoted by $R_{jklm}(\mathbf{x} - \mathbf{y}) := \mathbb{E}\{V_{jk}(\mathbf{x})V_{lm}(\mathbf{y})\}$, such that

$$R_{jklm}(\mathbf{x} - \mathbf{y}) = \int_{\mathbb{R}^3} e^{-i\mathbf{k} \cdot (\mathbf{x} - \mathbf{y})} S_{jklm}(\mathbf{k}) d\mathbf{k}. \quad (31)$$

$\mathbb{E}\{\cdot\}$ stands for mathematical expectation and $\mathbf{k} \mapsto S_{jklm}(\mathbf{k})$ are the spectral density functions of random matrix $(\mathbf{V}(\mathbf{x}), \mathbf{x} \in \mathbb{R}^3)$ for $j, k, l, m \in \{1, 2, \dots, n\}$. They satisfy the usual properties of spectral densities of homogeneous stochastic processes: $S_{jklm}(\mathbf{k}) = S_{lmjk}(\mathbf{k}) = S_{lmjk}(-\mathbf{k})$. Let $\mathbf{S}$ be the fourth-order spectral density tensor corresponding to the above spectral density functions. Thus, it can be shown that the coherence matrices $\mathbf{W}_{\alpha}$, $1 \leq \alpha \leq m$, satisfy coupled radiative transfer equations given by

$$\begin{aligned} \partial_t \mathbf{W}_{\alpha} + \{\omega_{\alpha}, \mathbf{W}_{\alpha}\} + [\mathbf{W}_{\alpha}, \mathbf{N}_{\alpha}] \\ = \sum_{\beta=1}^m \int_{\mathbb{R}^3} [\sigma_{\alpha\beta}(\mathbf{k}, \mathbf{k}') \mathbf{W}_{\beta}(\mathbf{k}')] \delta(\omega_{\alpha}(\mathbf{k}) - \omega_{\beta}(\mathbf{k}')) d\mathbf{k}' \\ - [\mathbf{S}_{\alpha}, \mathbf{W}_{\alpha}]. \end{aligned} \quad (32)$$

The differential scattering operators $\sigma_{\alpha\beta}(\mathbf{k}, \mathbf{k}')$ are defined by

$$[\sigma_{\alpha\beta}(\mathbf{k}, \mathbf{k}') \mathbf{W}_{\beta}(\mathbf{k}')] = 2\pi \omega_{\alpha}^2(\mathbf{k}) \times \mathbf{S}_{\alpha\beta}(\mathbf{k}, \mathbf{k}') \mathbf{W}_{\beta}(\mathbf{k}'), \quad (33)$$

where $\mathbf{S}_{\alpha\beta}(\mathbf{k}, \mathbf{k}')$ is a fourth-order tensor with elements

$$\begin{aligned} \mathbf{S}_{rstu}^{\alpha\beta}(\mathbf{k}, \mathbf{k}') = & \mathbf{c}_{\alpha_r}(\mathbf{k}) \otimes \mathbf{b}_{\beta_t}(\mathbf{k}') : \mathbf{S}(\mathbf{k} - \mathbf{k}') : \mathbf{c}_{\alpha_s}(\mathbf{k}) \\ & \otimes \mathbf{b}_{\beta_u}(\mathbf{k}'). \end{aligned} \quad (34)$$

The total scattering operators $\Sigma_{\alpha}(\mathbf{k})$ are defined by

$$\begin{aligned} \Sigma_{\alpha}(\mathbf{k}) = & \frac{1}{2} \sum_{\beta=1}^m \left\{ \int_{\mathbb{R}^3} [\sigma_{\alpha\beta}(\mathbf{k}, \mathbf{k}') \mathbf{I}_{R_{\beta}}] \delta(\omega_{\alpha}(\mathbf{k}) - \omega_{\beta}(\mathbf{k}')) d\mathbf{k}' \right. \\ & \left. - i \int_{\mathbb{R}^3} \frac{1}{\omega_{\alpha}(\mathbf{k}) - \omega_{\beta}(\mathbf{k}')} [\sigma_{\alpha\beta}(\mathbf{k}, \mathbf{k}') \mathbf{I}_{R_{\beta}}] d\mathbf{k}' \right\}. \end{aligned} \quad (35)$$

$\mathbf{b}_{\alpha_r}$ are the right eigenvectors of the mean dispersion matrix and $\mathbf{c}_{\alpha_r}$ are its left eigenvectors, given by $\mathbf{c}_{\alpha_r} = \mathbf{A} \mathbf{b}_{\alpha_r}$, for $1 \leq \alpha \leq m$ and $1 \leq r \leq R_{\alpha}$. The differential scattering operator $\sigma_{\alpha\beta}(\mathbf{x}, \mathbf{k}, \mathbf{k}')$ gives the rate at which the energy mode β with wave vector $\mathbf{k}'$ is converted to the energy mode α with wave vector $\mathbf{k}$, at position $\mathbf{x}$; the total scattering operator $\Sigma_{\alpha}(\mathbf{x}, \mathbf{k})$ gives the rate at which the energy mode α with wave vector $\mathbf{k}$ at position $\mathbf{x}$ is converted to all other modes or wave numbers (thus, it can account for intrinsic damping). The conservation of energy without dissipation is implied by the fact that one formally has $\sigma_{\alpha\beta}(\mathbf{k}, \mathbf{k}') \equiv \sigma_{\beta\alpha}(\mathbf{k}', \mathbf{k})$.

B. Radiative transfer equations for isotropic porous media

Let us consider a random perturbation matrix $(\mathbf{V}(\mathbf{x}), \mathbf{x} \in \mathbb{R}^3)$ having the form

$$\mathbf{V} = \begin{bmatrix} \tilde{\varrho}\mathbf{I}_3 & -\frac{\alpha}{\Phi}\tilde{F}\mathbf{I}_3 & 0 & 0 & 0 & 0 \\ -\frac{\Phi}{\alpha}\tilde{\varrho}\mathbf{I}_3 & \frac{\alpha\varrho}{\Phi\varrho f}\tilde{F}\mathbf{I}_3 & 0 & 0 & 0 & 0 \\ 0 & 0 & \tilde{\mu}\mathbf{I}_3 & 0 & 0 & 0 \\ 0 & 0 & 0 & \tilde{\mu}\mathbf{I}_3 & 0 & 0 \\ 0 & 0 & 0 & 0 & \tilde{\lambda} & -B\tilde{\lambda} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

where $(\tilde{\varrho}(\mathbf{x}), \mathbf{x} \in \mathbb{R}^3)$, $(\tilde{F}(\mathbf{x}), \mathbf{x} \in \mathbb{R}^3)$, $(\tilde{\mu}(\mathbf{x}), \mathbf{x} \in \mathbb{R}^3)$, and $(\tilde{\lambda}(\mathbf{x}), \mathbf{x} \in \mathbb{R}^3)$ are four real-valued, statistically homogeneous stochastic processes whose spectral measures are character-

ized by their power spectral density functions $\mathbf{k} \mapsto S_{\varrho}(\mathbf{k})$, $\mathbf{k} \mapsto S_F(\mathbf{k})$, $\mathbf{k} \mapsto S_{\mu}(\mathbf{k})$, and $\mathbf{k} \mapsto S_{\lambda}(\mathbf{k})$, respectively, and cross-spectral density functions $\mathbf{k} \mapsto S_{\varrho F}(\mathbf{k})$, $\mathbf{k} \mapsto S_{\varrho \mu}(\mathbf{k})$, $\mathbf{k} \mapsto S_{\varrho \lambda}(\mathbf{k})$, $\mathbf{k} \mapsto S_{F \mu}(\mathbf{k})$, $\mathbf{k} \mapsto S_{F \lambda}(\mathbf{k})$, and $\mathbf{k} \mapsto S_{\lambda \mu}(\mathbf{k})$. This model of perturbations corresponds to random fluctuations of the mean density ϱ , the mean inverse formation factor F^{-1} (defined by $F = \alpha/\Phi$), and the mean inverse Lamé's moduli μ_0^{-1} and λ_0^{-1} . Therefore, the differential scattering operator for Pj to Pk mode conversion, $j \neq k \in \{1, 2\}$, is a simple multiplicative operator

$$[\sigma_{\text{PjPk}}(\mathbf{k}, \mathbf{k}') w_{\text{Pk}}(\mathbf{k}')] = \sigma_{\text{pp}}(\mathbf{k}, \mathbf{k}') w_k(\mathbf{k}'), \quad (36)$$

characterized by the (scalar) scattering cross section

$$\begin{aligned} \sigma_{\text{pp}}(\mathbf{k}, \mathbf{k}') = \frac{\pi}{2} c_{\text{p}}^2 |\mathbf{k}| |\mathbf{k}'| & \left[\frac{c_2}{c_{\text{p}}} (\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')^2 (S_{\varrho}(\mathbf{k} - \mathbf{k}') + S_F(\mathbf{k} - \mathbf{k}') + 2 \operatorname{Re} S_{\varrho F}(\mathbf{k} - \mathbf{k}')) + 4 \frac{c_{\text{S}}^4}{c_{\text{p}}^3 c_2} (\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')^4 S_{\mu}(\mathbf{k} - \mathbf{k}') \right. \\ & + 4 \frac{c_{\text{S}}^2}{c_{\text{p}}^2} (\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')^3 (\operatorname{Re} S_{\varrho \mu}(\mathbf{k} - \mathbf{k}') + \operatorname{Re} S_{F \mu}(\mathbf{k} - \mathbf{k}')) + 2 \left(1 - 2 \frac{c_{\text{S}}^2}{c_{\text{p}}^2} \right) (\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') (\operatorname{Re} S_{\varrho \lambda}(\mathbf{k} - \mathbf{k}')) \\ & \left. + \operatorname{Re} S_{F \lambda}(\mathbf{k} - \mathbf{k}') + \frac{4 c_{\text{S}}^2}{c_{\text{p}} c_2} \left(1 - 2 \frac{c_{\text{S}}^2}{c_{\text{p}}^2} \right) (\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')^2 \operatorname{Re} S_{\lambda \mu}(\mathbf{k} - \mathbf{k}') + \frac{c_{\text{p}}}{c_2} \left(1 - 2 \frac{c_{\text{S}}^2}{c_{\text{p}}^2} \right)^2 S_{\lambda}(\mathbf{k} - \mathbf{k}') \right]. \end{aligned} \quad (37)$$

The differential scattering operator for Pj scattering without mode conversion, $j \in \{1, 2\}$, is also a multiplicative operator

$$[\sigma_{\text{PjPj}}(\mathbf{k}, \mathbf{k}') w_{\text{Pj}}(\mathbf{k}')] = \sigma_{\text{pp}}^j(\mathbf{k}, \mathbf{k}') w_{\text{Pj}}(\mathbf{k}'), \quad (38)$$

characterized by the scattering cross section

$$\begin{aligned} \sigma_{\text{pp}}^j(\mathbf{k}, \mathbf{k}') = \frac{\pi}{2} c_{\text{pj}}^2 |\mathbf{k}|^2 & \left[(\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')^2 (S_{\varrho}(\mathbf{k} - \mathbf{k}') + S_F(\mathbf{k} - \mathbf{k}') + 2 \operatorname{Re} S_{\varrho F}(\mathbf{k} - \mathbf{k}')) + 4 \frac{c_{\text{S}}^4}{c_{\text{pj}}^4} (\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')^4 S_{\mu}(\mathbf{k} - \mathbf{k}') \right. \\ & + 4 \frac{c_{\text{S}}^2}{c_{\text{pj}}^2} (\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')^3 (\operatorname{Re} S_{\varrho \mu}(\mathbf{k} - \mathbf{k}') + \operatorname{Re} S_{F \mu}(\mathbf{k} - \mathbf{k}')) + 2 \frac{c_{\text{p}}^2}{c_{\text{pj}}^2} \left(1 - 2 \frac{c_{\text{S}}^2}{c_{\text{p}}^2} \right) (\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') (\operatorname{Re} S_{\varrho \lambda}(\mathbf{k} - \mathbf{k}')) \\ & \left. + \operatorname{Re} S_{F \lambda}(\mathbf{k} - \mathbf{k}') + 4 \frac{c_{\text{S}}^2 c_{\text{p}}^2}{c_{\text{pj}}^4} \left(1 - 2 \frac{c_{\text{S}}^2}{c_{\text{p}}^2} \right) (\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')^2 \operatorname{Re} S_{\lambda \mu}(\mathbf{k} - \mathbf{k}') + \frac{c_{\text{p}}^4}{c_{\text{pj}}^4} \left(1 - 2 \frac{c_{\text{S}}^2}{c_{\text{p}}^2} \right)^2 S_{\lambda}(\mathbf{k} - \mathbf{k}') \right]. \end{aligned} \quad (39)$$

The differential scattering operator for Pj to S mode conversion, $j \in \{1, 2\}$, is given by

$$[\sigma_{\text{PjS}}(\mathbf{k}, \mathbf{k}') \mathbf{W}_{\text{S}}(\mathbf{k}')] = \sigma_{\text{ps}}(\mathbf{k}, \mathbf{k}') \mathbf{G}(\mathbf{k}, \mathbf{k}') : \mathbf{W}_{\text{S}}(\mathbf{k}'), \quad (40)$$

where

$$\begin{aligned} \sigma_{\text{ps}}(\mathbf{k}, \mathbf{k}') = \frac{\pi}{2} c_{\text{S}}^2 [|\mathbf{k}'|^2 (S_{\varrho}(\mathbf{k} - \mathbf{k}') + S_F(\mathbf{k} - \mathbf{k}')) & + 2 \operatorname{Re} S_{\varrho F}(\mathbf{k} - \mathbf{k}') + 4 |\mathbf{k}|^2 (\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')^2 S_{\mu}(\mathbf{k} - \mathbf{k}') \\ & - \mathbf{k}') + 4 |\mathbf{k}| |\mathbf{k}'| (\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') (\operatorname{Re} S_{\varrho \mu}(\mathbf{k} - \mathbf{k}')) \\ & + \operatorname{Re} S_{F \mu}(\mathbf{k} - \mathbf{k}'))], \end{aligned} \quad (41)$$

and $\mathbf{G}(\mathbf{k}, \mathbf{k}')$ is the 2×2 real matrix defined by $G_{rs}(\mathbf{k}, \mathbf{k}') := (\hat{\mathbf{k}} \cdot \hat{\mathbf{z}}_r(\mathbf{k}')) (\hat{\mathbf{k}} \cdot \hat{\mathbf{z}}_s(\mathbf{k}'))$. Conversely, the differential scatter-

ing operator for S to Pj mode conversion, $j \in \{1, 2\}$, is

$$[\sigma_{\text{SPj}}(\mathbf{k}, \mathbf{k}') w_{\text{Pj}}(\mathbf{k}')] = \sigma_{\text{ps}}(\mathbf{k}', \mathbf{k}) \mathbf{G}(\mathbf{k}', \mathbf{k}) w_{\text{Pj}}(\mathbf{k}'). \quad (42)$$

Finally, the differential scattering operator for S scattering without mode conversion is given by

$$\begin{aligned} [\sigma_{\text{SS}}(\mathbf{k}, \mathbf{k}') \mathbf{W}_{\text{S}}(\mathbf{k}')] = \sigma_{\text{ss}}^{\text{TT}}(\mathbf{k}, \mathbf{k}') \mathbf{T}(\mathbf{k}, \mathbf{k}') \mathbf{W}_{\text{S}}(\mathbf{k}') \mathbf{T}(\mathbf{k}', \mathbf{k}) & + \sigma_{\text{ss}}^{\text{I}\Gamma}(\mathbf{k}, \mathbf{k}') \Gamma(\mathbf{k}, \mathbf{k}') \mathbf{W}_{\text{S}}(\mathbf{k}') \Gamma(\mathbf{k}', \mathbf{k}) \\ & + \sigma_{\text{ss}}^{\text{TT}}(\mathbf{k}, \mathbf{k}') \mathbf{T}(\mathbf{k}, \mathbf{k}') \mathbf{W}_{\text{S}}(\mathbf{k}') \Gamma(\mathbf{k}', \mathbf{k}) \\ & + \sigma_{\text{ss}}^{\text{TI}}(\mathbf{k}', \mathbf{k}) \Gamma(\mathbf{k}, \mathbf{k}') \mathbf{W}_{\text{S}}(\mathbf{k}') \mathbf{T}(\mathbf{k}', \mathbf{k}), \end{aligned} \quad (43)$$

where

$$\begin{aligned}
\sigma_{ss}^{TT}(\mathbf{k}, \mathbf{k}') &= \frac{\pi}{2} c_s^2 |\mathbf{k}|^2 (S_\varrho(\mathbf{k} - \mathbf{k}') + S_F(\mathbf{k} - \mathbf{k}')) \\
&\quad + 2 \operatorname{Re} S_{\varrho F}(\mathbf{k} - \mathbf{k}'), \\
\sigma_{ss}^{\Gamma\Gamma}(\mathbf{k}, \mathbf{k}') &= \frac{\pi}{2} c_s^2 |\mathbf{k}|^2 S_\mu(\mathbf{k} - \mathbf{k}'), \\
\sigma_{ss}^{T\Gamma}(\mathbf{k}, \mathbf{k}') &= \frac{\pi}{2} c_s^2 |\mathbf{k}|^2 (\operatorname{Re} S_{\varrho\mu}(\mathbf{k} - \mathbf{k}')) \\
&\quad + \operatorname{Re} S_{F\mu}(\mathbf{k} - \mathbf{k}')).
\end{aligned} \tag{44}$$

$\mathbf{T}(\mathbf{k}, \mathbf{k}')$ and $\mathbf{\Gamma}(\mathbf{k}, \mathbf{k}')$ are the 2×2 real matrices defined by $T_{rs}(\mathbf{k}, \mathbf{k}') := \hat{\mathbf{z}}_r(\mathbf{k}) \cdot \hat{\mathbf{z}}_s(\mathbf{k}')$ and $\Gamma_{rs}(\mathbf{k}, \mathbf{k}') := (\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') (\hat{\mathbf{z}}_r(\mathbf{k}) \cdot \hat{\mathbf{z}}_s(\mathbf{k}') + (\hat{\mathbf{k}} \cdot \hat{\mathbf{z}}_s(\mathbf{k}')) (\hat{\mathbf{k}}' \cdot \hat{\mathbf{z}}_r(\mathbf{k})))$. These results show that the forward (“+”) and backward (“−”) waves are uncoupled in the random case.

V. DIFFUSION LIMIT

It is now well established that in the diffusion limit, a heat equation can be recovered from a radiative transfer equation; see for example Ref. 23 for scalar waves or Ref. 5 for polarized waves. This regime is reached after long propagation distances with respect to the scattering mean-free path of the medium, the latter being the mean distance traveled by waves before their direction of propagation is significantly altered. Waves in this regime have lost their initial direction of propagation as well as their initial polarization. From such a diffusion equation, one can also obtain proportionality relationships between the energy densities of particular propagating modes and their associated velocities, independent of the scattering mechanism. The scattering mean-free path is defined for a particular propagating mode α by

$$\ell_\alpha(\mathbf{x}, \mathbf{k}) = \frac{|\nabla_{\mathbf{k}} \omega_\alpha|}{\sum_\alpha(\mathbf{x}, \mathbf{k})}. \tag{45}$$

The derivation of the radiative transfer equations has been carried out so far for high frequencies such that $\Lambda \ll c_\alpha t$ and it remains applicable as long as $\Lambda \ll \ell_\alpha$, where Λ is a typical wavelength. The diffusion limit considers that propagation distances are also much longer than the scattering mean-free path: $c_\alpha t \gg \ell_\alpha$. Parameter fluctuations are also assumed to be statistically isotropic. This implies that the differential scattering cross sections are rotationally invariant inasmuch as they depend on $|\mathbf{k}|$ and $\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}'$ solely. The former may be written under these conditions $\sigma_{\alpha\beta}(\mathbf{k}, \mathbf{k}') := c_\beta |\mathbf{k}'|^{-2} \times \tilde{\sigma}_{\alpha\beta}(|\mathbf{k}|, |\mathbf{k}'|, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')$ and the total scattering cross sections, depending on $|\mathbf{k}|$ solely, are

$$\begin{aligned}
\Sigma_{\alpha\beta}(|\mathbf{k}|) &= \frac{R_\beta}{R_\alpha} \left(\frac{c_\alpha}{c_\beta} \right)^3 \Sigma_{\beta\alpha} \left(\frac{c_\alpha}{c_\beta} |\mathbf{k}| \right) \\
&= \int_{S^2} \tilde{\sigma}_{\alpha\beta} \left(|\mathbf{k}|, \frac{c_\alpha}{c_\beta} |\mathbf{k}|, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}' \right) d\hat{\mathbf{k}}',
\end{aligned}$$

where $d\hat{\mathbf{k}}$ is the elementary solid angle of $\mathbb{R}^3$ and S^2 is its unit sphere.

Based on these assumptions, Eq. (32) yields the following diffusive regime for the present model of porous media. Coherence matrix $\mathbf{W}_S^+$ and specific intensities w_{pj}^+ become independent of $\hat{\mathbf{k}}$ and have the form

$$\mathbf{W}_S^+(\mathbf{x}, \mathbf{k}, t) = \psi(\mathbf{x}, |\mathbf{k}|, t) \mathbf{I}_2, \tag{46a}$$

$$w_{pj}^+(\mathbf{x}, \mathbf{k}, t) = \psi \left(\mathbf{x}, \frac{c_{pj}}{c_S} |\mathbf{k}|, t \right), \quad j \in \{1, 2\}, \tag{46b}$$

where $\psi(\mathbf{x}, |\mathbf{k}|, t)$ is a scalar function satisfying the following diffusion equation:

$$\partial_t \psi(\mathbf{x}, |\mathbf{k}|, t) = \nabla_{\mathbf{x}} \cdot [D(|\mathbf{k}|) \nabla_{\mathbf{x}} \psi(\mathbf{x}, |\mathbf{k}|, t)]. \tag{47}$$

The diffusion constant D is

$$D(|\mathbf{k}|) = \frac{2D_S(|\mathbf{k}|) + \left(\frac{c_S}{c_{P1}} \right)^3 D_{P1} \left(\frac{c_S}{c_{P1}} |\mathbf{k}| \right) + \left(\frac{c_S}{c_{P2}} \right)^3 D_{P2} \left(\frac{c_S}{c_{P2}} |\mathbf{k}| \right)}{2 + \left(\frac{c_S}{c_{P1}} \right)^3 + \left(\frac{c_S}{c_{P2}} \right)^3}, \tag{48}$$

where the $D_\alpha(|\mathbf{k}|)$'s are partial diffusion coefficients which have the usual expression $D_\alpha = \frac{1}{3} c_\alpha \ell_\alpha^*$, $\alpha \in \{S, P1, P2\}$. The diffusion constant characterizes the growth speed of the diffusive halo in a disordered medium. The so-called transport mean-free paths ℓ_α^* satisfy

$$\mathbf{T}^{-1}(|\mathbf{k}|) \mathbf{\Lambda} \star (|\mathbf{k}|) = \mathbf{C}, \tag{49}$$

where $\mathbf{\Lambda} \star (|\mathbf{k}|) = (\ell_S^*(|\mathbf{k}|), \ell_{P1}^*(c_S/c_{P1}|\mathbf{k}|), \ell_{P2}^*(c_S/c_{P2}|\mathbf{k}|))$, $\mathbf{C} = (c_S, c_{P1}, c_{P2})$, and

$$\mathbf{T}^{-1}(|\mathbf{k}|) = \begin{bmatrix} \Sigma_S(|\mathbf{k}|) - \Sigma_{SS}^*(|\mathbf{k}|) & -\Sigma_{SP1}^*(|\mathbf{k}|) & -\Sigma_{SP2}^*(|\mathbf{k}|) \\ -\Sigma_{P1S}^* \left(\frac{c_S}{c_{P1}} |\mathbf{k}| \right) & \Sigma_{P1} \left(\frac{c_S}{c_{P1}} |\mathbf{k}| \right) - \Sigma_{P1P1}^* \left(\frac{c_S}{c_{P1}} |\mathbf{k}| \right) & -\Sigma_{P1P2}^* \left(\frac{c_S}{c_{P1}} |\mathbf{k}| \right) \\ -\Sigma_{P2S}^* \left(\frac{c_S}{c_{P2}} |\mathbf{k}| \right) & -\Sigma_{P2P1}^* \left(\frac{c_S}{c_{P2}} |\mathbf{k}| \right) & \Sigma_{P2} \left(\frac{c_S}{c_{P2}} |\mathbf{k}| \right) - \Sigma_{P2P2}^* \left(\frac{c_S}{c_{P2}} |\mathbf{k}| \right) \end{bmatrix},$$

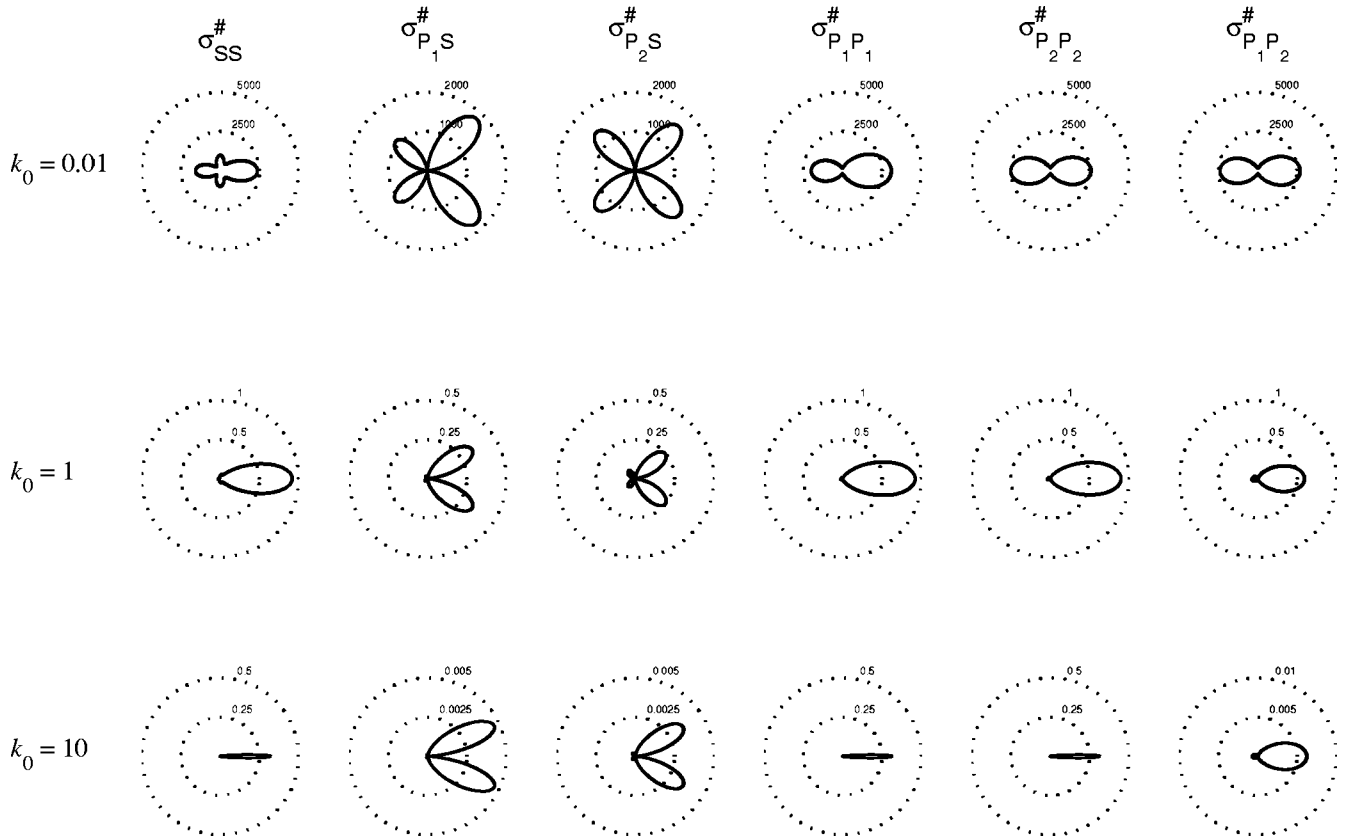


FIG. 1. Normalized scattering cross sections $\sigma_{\alpha\beta}^{\#}$, $\alpha, \beta \in \{S, P_1, P_2\}$, as functions of the angle Θ between $\mathbf{k}$ and $\mathbf{k}'$, for various values of the dimensionless wave number $k_0 = \gamma^{-1}|\mathbf{k}|$, and for a porous medium with Gaussian correlations of perturbations; $h_{\varrho} = h_F = 0.1$ and $h_{\lambda} = h_{\mu} = 1$.

with

$$\begin{aligned} \Sigma_{\alpha\beta}^*(|\mathbf{k}|) &= \frac{R_{\beta}}{R_{\alpha}} \left(\frac{c_{\alpha}}{c_{\beta}} \right)^3 \Sigma_{\beta\alpha}^* \left(\frac{c_{\alpha}}{c_{\beta}} |\mathbf{k}| \right) \\ &= \int_{S^2} \tilde{\sigma}_{\alpha\beta} \left(|\mathbf{k}|, \frac{c_{\alpha}}{c_{\beta}} |\mathbf{k}|, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}' \right) \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}' d\hat{\mathbf{k}}', \end{aligned}$$

for $\alpha, \beta \in \{S, P_1, P_2\}$. These coefficients characterize the tendency of total scattering to be in the forward direction. It is worth noting at this stage that analytical expressions of the diffusion parameters like the mean-free paths or the diffusion coefficients are useful because these quantities can be estimated from experimental data while correlation lengths and standard deviations of parameter fluctuations are usually not observable; see for example Refs. 24–28 for such experimental analyses.

At last, the space-time energy densities $\mathcal{E}_S(\mathbf{x}, t) = \int_{\mathbb{R}^3} \text{Tr} \mathbf{W}_S^+(\mathbf{x}, \mathbf{k}, t) d\mathbf{k}$ and $\mathcal{E}_{P_j}(\mathbf{x}, t) = \int_{\mathbb{R}^3} \mathbf{w}_{P_j}^+(\mathbf{x}, \mathbf{k}, t) d\mathbf{k}$, for $j \in \{1, 2\}$, satisfy

$$\frac{\mathcal{E}_{P_1}(\mathbf{x}, t)}{\mathcal{E}_{P_2}(\mathbf{x}, t)} = \left(\frac{c_{P_2}}{c_{P_1}} \right)^3, \quad \frac{\mathcal{E}_S(\mathbf{x}, t)}{\mathcal{E}_{P_j}(\mathbf{x}, t)} = 2 \left(\frac{c_{P_j}}{c_S} \right)^3, \quad (50)$$

in the diffusion limit, whatever the underlying (random) heterogeneity of the medium or the source may be. The above equipartition rules are indicators of the relevance of the diffusion approximation: the latter is acceptable as soon as these relations are fulfilled by the energy densities, at each location $\mathbf{x}$ of the medium and at each time t . Furthermore, they show that in this regime, most of the energy in the

skeleton is transported by the S modes, no matter how it was distributed initially, because $c_{P_2} < c_S < c_{P_1}$. This characteristic feature can be said to be “universal” because, again, it does not depend on the details of the heterogeneities and/or the source, and it is observed everywhere in the medium at any time independent of any averaging process.

VI. NUMERICAL EXAMPLE

The compressional wave speeds for energy transport are computed by Eq. (24). One can use for example the data provided in Ref. 29 for the experiments carried on in Ref. 30 for a water-saturated porous solid composed of sintered glass spheres: $\varrho_f = 1000 \text{ kg/m}^3$, $\varrho_s = 2480 \text{ kg/m}^3$, $K_f = 2.25 \cdot 10^9 \text{ Pa}$, $K_b = 15.2 \cdot 10^9 \text{ Pa}$, $K_s = 40.7 \cdot 10^9 \text{ Pa}$, $\mu_0 = 10.75 \cdot 10^9 \text{ Pa}$, $\Phi = 0.283$, and $\alpha = 1.82$. One obtains $c_{P_1} = 4053 \text{ m/s}$ and $c_{P_2} = 1045 \text{ m/s}$, which compares remarkably well with the experimental estimations $c_{P_1} \simeq 4050 \text{ m/s}$ and $c_{P_2} \simeq 1040 \text{ m/s}$.³⁰ For the diffusion parameters, a statistically isotropic, Gaussian model of correlation of the fluctuations of ϱ , F^{-1} , μ_0^{-1} , and λ_0^{-1} is considered

$$S_{ab}(|\mathbf{k}|) = \frac{h_a h_b r_{ab}}{8 \sqrt{\pi^3} \gamma^3} \exp \left(- \frac{|\mathbf{k}|^2}{4 \gamma^2} \right), \quad a, b \in \{\varrho, F, \mu, \lambda\}, \quad (51)$$

where γ^{-1} is the correlation length of these parameters. The h_a 's, $a, b \in \{\varrho, F, \mu, \lambda\}$, are their fluctuation amplitudes, and the r_{ab} 's are coherency coefficients such that $|r_{ab}| \leq 1$ and $r_{aa} = 1$. The normalized scattering cross sections

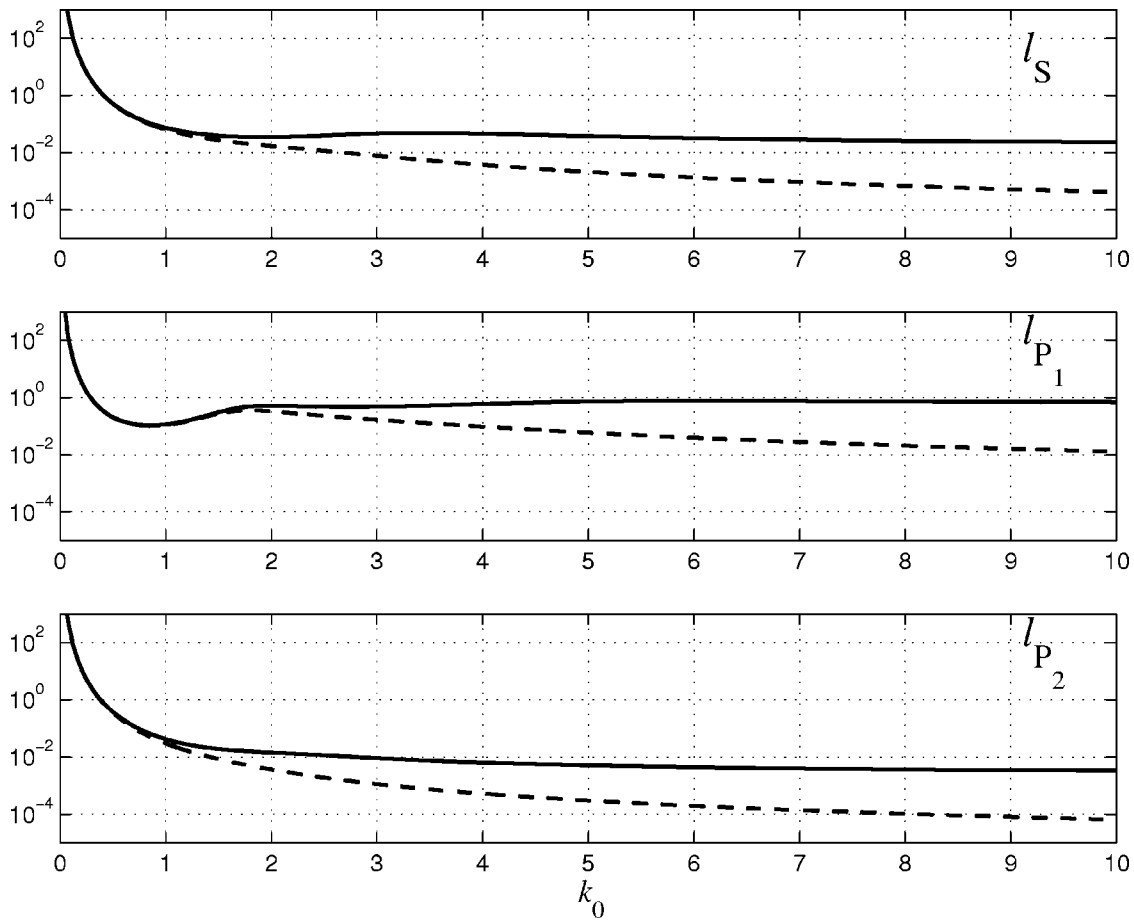


FIG. 2. Normalized scattering (dashed lines) and transport (solid lines) mean-free paths $\gamma\ell_\alpha$ and $\gamma\ell_\alpha^*$, $\alpha=S, P_1$ or P_2 , as functions of the dimensionless wave number $k_0 = \gamma^{-1}|\mathbf{k}|$, for a porous medium with Gaussian correlations of perturbations; $h_e = h_F = 0.1$ and $h_\lambda = h_\mu = 1$.

$$\sigma_{\alpha\beta}^\#(|\mathbf{k}|, \Theta) = \left(\frac{c_\alpha}{c_\beta}\right)^3 \frac{\sigma_{\alpha\beta}(\mathbf{k}, \mathbf{k}')}{\gamma c_\alpha \Sigma_{\alpha\beta}(|\mathbf{k}|)}, \quad \alpha, \beta \in \{S, P_1, P_2\},$$

where $\cos \Theta = \hat{\mathbf{k}} \cdot \hat{\mathbf{k}'}$, are plotted in polar coordinates in Fig. 1 as functions of Θ for various values of the dimensionless wave number $k_0 := \gamma^{-1}|\mathbf{k}|$, and for the above data with $h_e = h_F = 0.1$, $h_\mu = h_\lambda = 1$ (strongly fluctuating inverse Lamé's moduli with respect to the density and inverse formation factor), $r_{eF} = r_{\mu\lambda} = 0.8$, and all other coherency coefficients being equal to 0.2. We outline their overall strong anisotropy—even at low frequencies—despite the assumption of statistical isotropy. Also, forward scattering dominates as the frequency (wave number) increases. Figure 2 displays the normalized scattering mean-free paths $\gamma\ell_\alpha$ (dashed lines) and the normalized transport mean-free paths $\gamma\ell_\alpha^*$ (solid lines), $\alpha=S, P_1$, or P_2 , as functions of the dimensionless wave number. Transport mean-free paths get larger than the scattering mean-free paths as frequency increases, as expected from the increased anisotropy of scattering in the forward direction usually observed as frequency increases. One also notes that the former have nonzero, frequency-independent values over a broad frequency range: this is the so-called Rayleigh–Gans regime. Figure 3 displays the normalized diffusion constant $\gamma D/c_S$ as a function of the dimensionless wave number k_0 , for the same data and set of fluctuation amplitudes and coherency coefficients. The fact that these diffusion parameters have very low asymptotic

values indicates that weak localization may dominate in porous media at the higher frequencies. That is the reason why such materials are used for acoustic insulation, among other possible applications. Finally, the author wishes to point out that choosing different values for the fluctuation amplitudes and coherency coefficients does not modify the diffusion parameters significantly, both from a quantitative and a qualitative point of view.

VII. CONCLUSIONS

In this work the acoustic diffusion properties of a mechanically impacted fluid-saturated poro-visco-elastic medium in the higher frequency range of vibration have been obtained. They are associated with the diffusion equation which must be satisfied by the energy density (in space and time) at late times or after long propagation distances in a randomly perturbed medium. This diffusive regime is also characterized by universal equipartition rules between the modes of propagation, S shear waves of the bulk solid phase, and P1 and P2 compressional waves of the solid and fluid phases. In the course of the derivation one has also obtained analytical expressions of the associated energy speeds and observed that they are nondispersive in the higher frequency range of vibration.

One important issue in the proposed theory remains unsolved: what are the boundary conditions for the energy den-

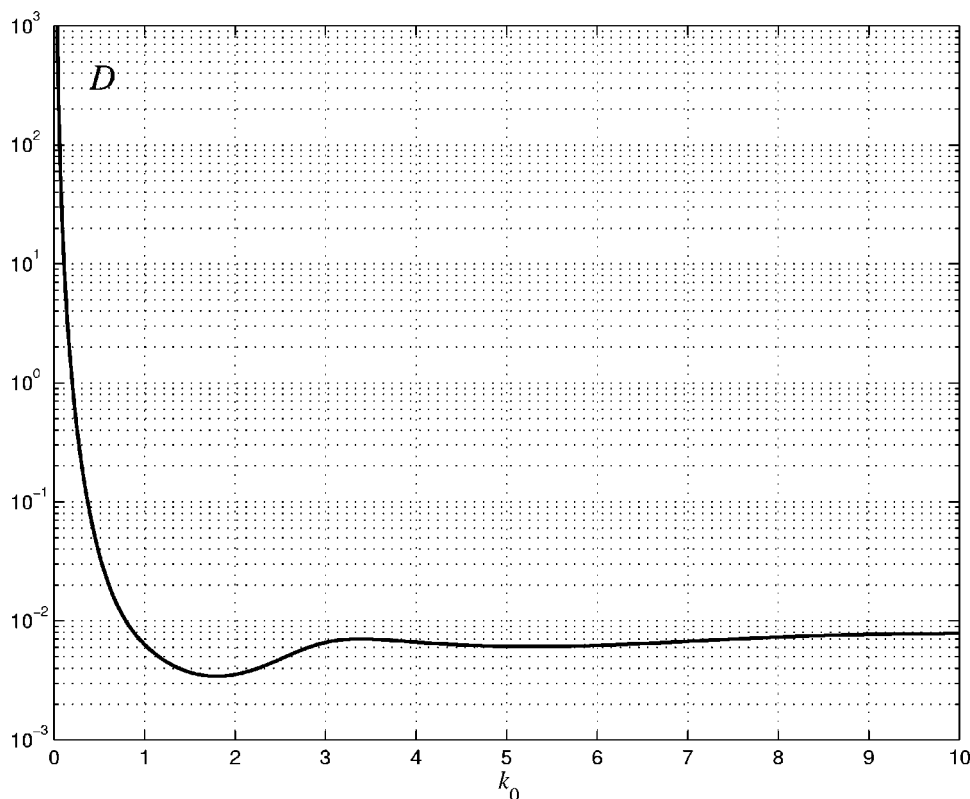


FIG. 3. Normalized diffusion coefficient $\gamma D/c_s$ as a function of the dimensionless wave number $k_0 = \gamma^{-1}|\mathbf{k}|$, for a porous medium with Gaussian correlations of perturbations; $h_e = h_F = 0.1$ and $h_\lambda = h_\mu = 1$.

sity with respect to the boundary conditions imposed to the original displacement field of the medium? Further theoretical work on this approach shall be devoted to this aspect of the model. Also, some work shall be done to possibly link the above results to the acoustic insulation properties of porous materials, such as plastic or metallic foams. One has not explored that possibility yet, though it may have some important applications in the aerospace or automotive industry.

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